第2章 电路的等效变换

2.4 电源的等效变换

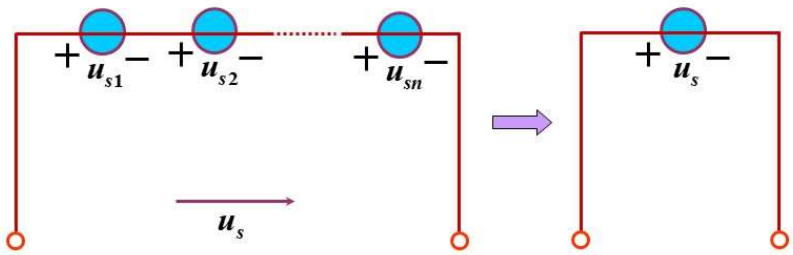
2.5 实际电源的两种模型及其等效变换

2.6 输入电阻

第一次习题课

2.4 电压源、电流源的串联和并联

一、电压源的串联



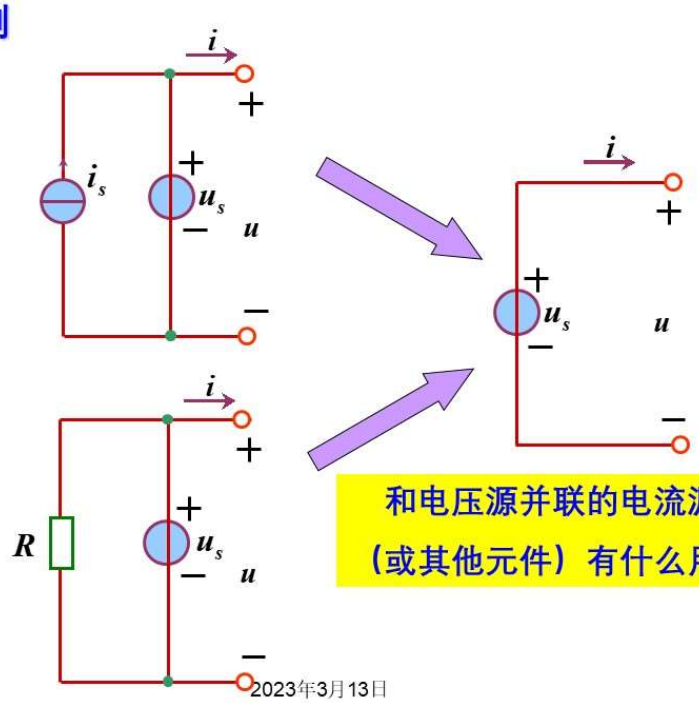
根据KVL可得：$u_s = u_{s1} + u_{s2} + \ldots\ldots + u_{sn}$

$$= \sum_{k=1}^{n} u_{sk}$$

（外特性不变）。

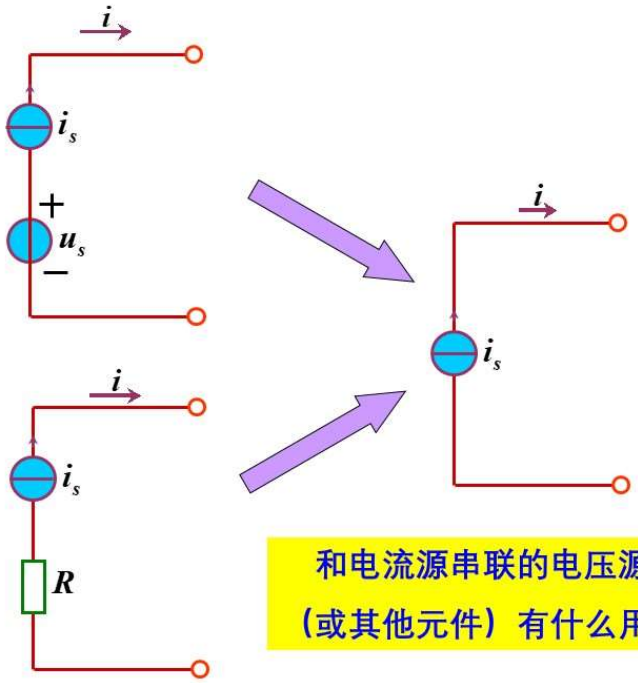
注意：只有相同的电压源才能允许并联。否则，烧毁电压表。

特例



和电压源并联的电流源
（或其他元件）有什么用？

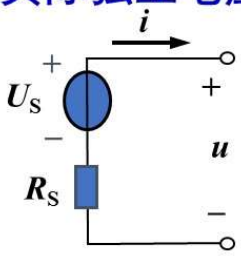
2023年3月13日



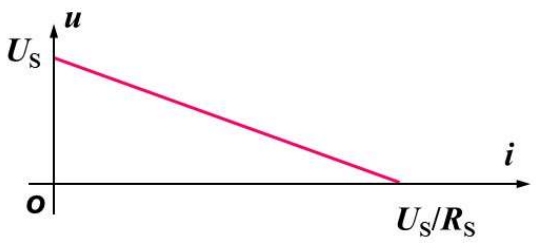
和电流源串联的电压源
（或其他元件）有什么用？

2.5 实际独立源的等效变换

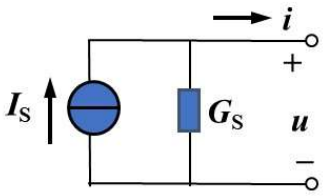
一、实际独立电压源



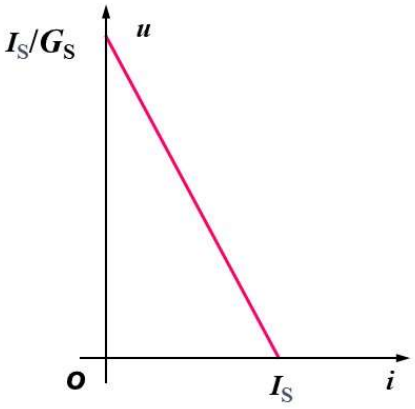
$$u=U_S - R_S\, i$$



二、 实际独立电流源

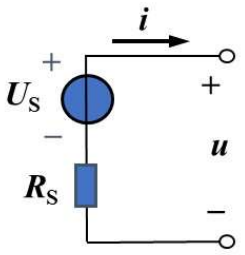


$$i=I_S - G_S\, u$$

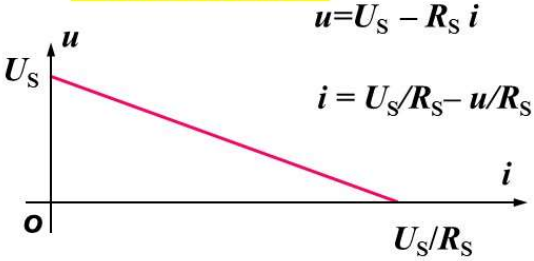


三、电源等效变换

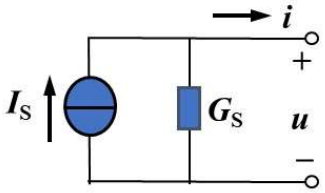
二者如何等效?

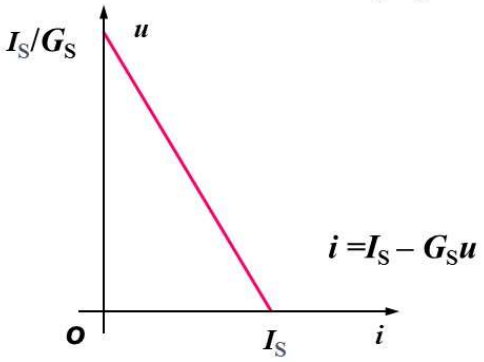


$u=U_S - R_S\, i$



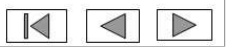
$i = U_S/R_S - u/R_S$





$i = I_S - G_S u$

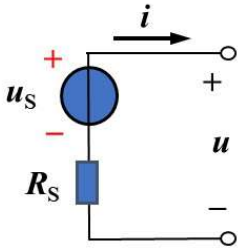
$I_S=U_S/R_S$ ， $G_S=1/R_S$



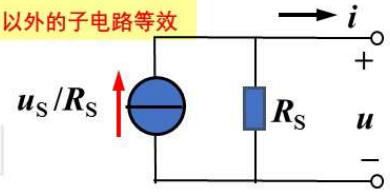
从电压源变换为电流源

等效的相对性

只对端口以外的子电路等效

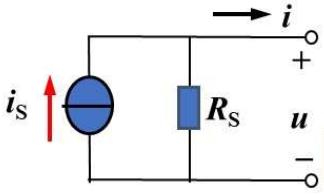


$I_S=U_S/R_S$

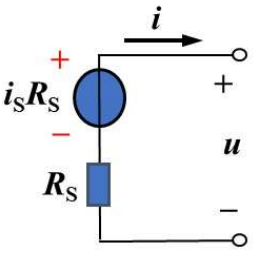


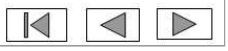
注意参考方向

从电流源变换为电压源



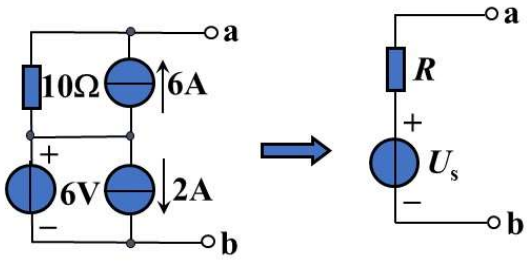
$U_S=I_S R_S$





单选题 1分

U_s= _____V



A 6

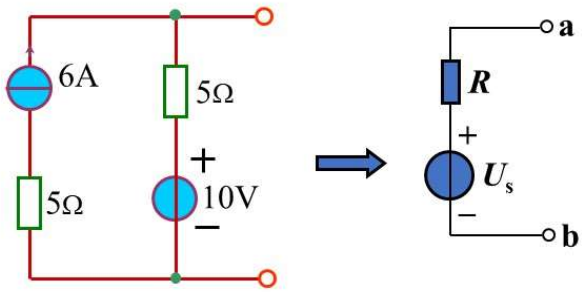
B 60

C −60

D 66

单选题 1分

U_s= _____V



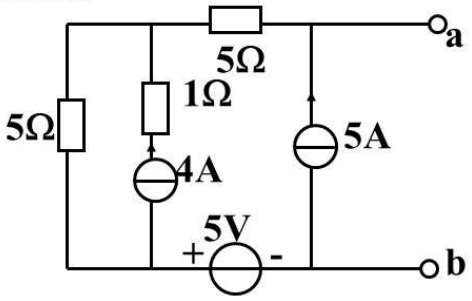
A 20

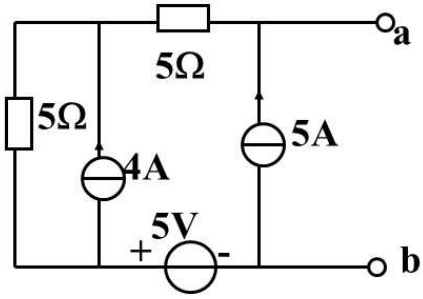
B 40

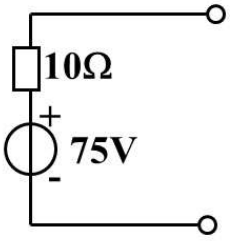
C 60

D -40

Ex 1：求等效电路



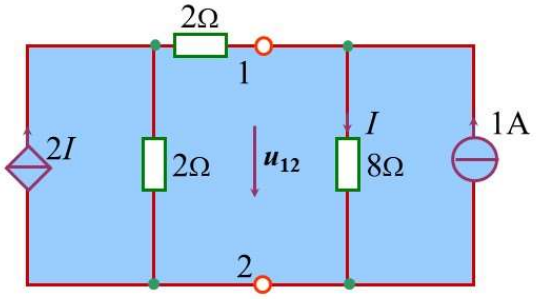


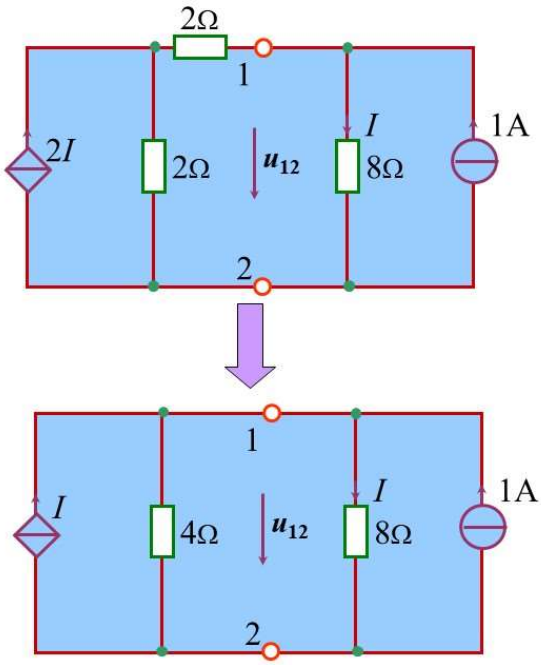


2、特殊情况（含受控源）

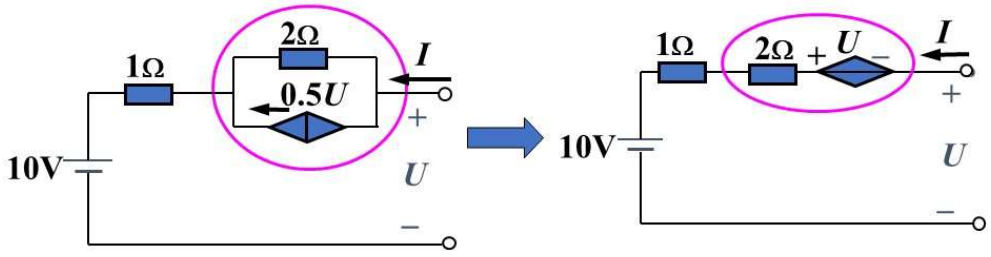
受控电压源、电组的串联组合与受控电流源、电导的并联可以用上述同样的方法进行变换。此时应把受控源当作独立电源来处理。注意：在等效变换过程中控制量必须在电路中保持完整的形式。

例2.6：试用电源的等效变换方法求下图所示电路中的电压u_{12}。

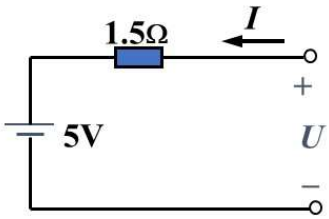




例2　求端口*U-I*关系和最简等效电路。



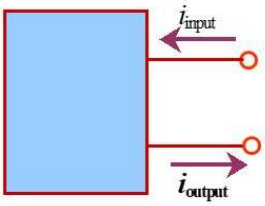
$$2U = 3I + 10$$



2.6 输入电阻

一、一端口网络（二端子网络）

定义：一个网络具有两个引出端子与外电路相联而不管其内部结构如何复杂，这样的网络叫**一端口网络。**



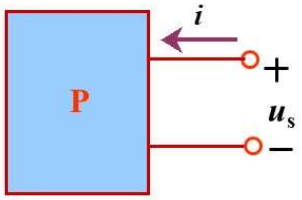
$$i_{input} = i_{output}$$

一端口网络分为含源一端口网络与无源一端口网络。

二、输入电阻（入端电阻，针对无源一端口网络而言）

输入电阻（R_{in}）：对无源一端口网络外施电压源u_s或电流源i_s产生相应电流i或电压u，则：

$$R_{in} = \frac{u_s}{i} \quad 或 \quad R_{in} = \frac{u}{i_s}$$



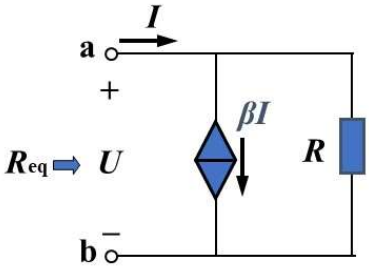
注意：无源一端口网络内部可含电阻和受控源。

三、等效电阻R_{eq}

定义：等效替代无源一端口网络的电阻。

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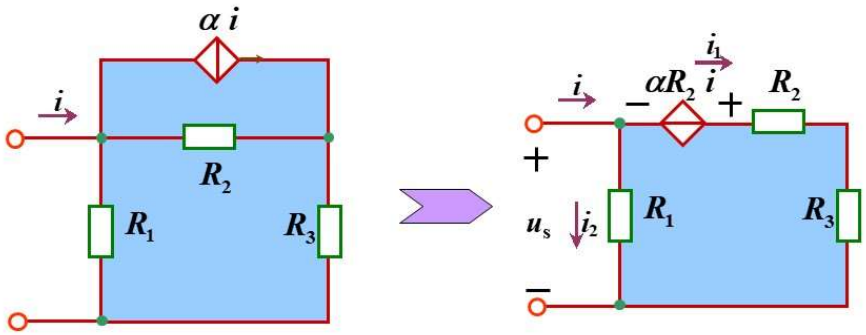
求入端等效电阻 → 求端口上的电压电流关系 → 加压求流或加流求压



$$I=\beta I+\frac{U}{R} \longrightarrow U=R(1-\beta)I$$

$$R_{eq}=U/I=(1-\beta)R$$

例2.7：求下图所示一端口的输入电阻。



$$u_s=-\alpha R_2 i+(R_2+R_3)i_1 \quad (i_1=i-i_2=i-\frac{u_s}{R_1})$$

$$=-\alpha R_2 i+(R_2+R_3)(i-\frac{u_s}{R_1})$$

$$R_{in}=\frac{u_s}{i}=\frac{R_1R_3+(1-\alpha)R_1R_2}{R_1+R_2+R_3}$$

总结：如何求二端网络的入端电阻

串并联

平衡电桥

Δ—Y变换

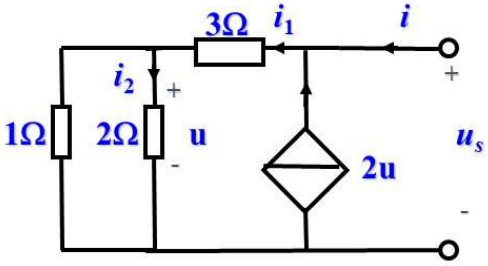
↓ 电阻二端网络求解顺序

加压求流/

加流求压

← 含受控源二端网络入端电阻

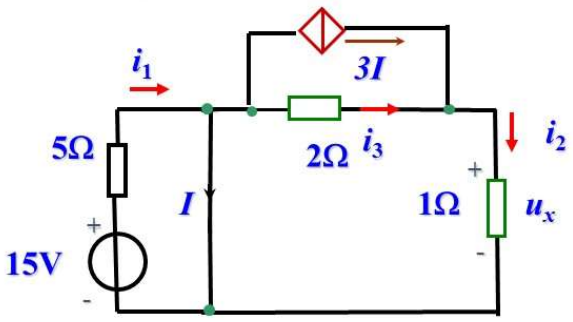
例：求输入电阻R



$$\left.\begin{aligned} u_s &= 3i_1 + u \\ i &= i_1 - 2u \\ u &= 2i_2 = 2\frac{1}{1+2}i_1 \end{aligned}\right\} \Rightarrow R = \frac{u_s}{i} = -11\Omega$$

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Ex：求支路电压u_x

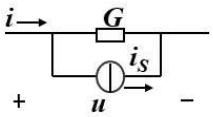


$$15+5i_1=0$$

$$\left.\begin{aligned} i_2 &= i_3 + 3I \\ i_2 + I &= i_1 \\ 2i_3 + i_2 &= 0 \end{aligned}\right\} \Rightarrow u_x = i_2 = 2\text{V}$$

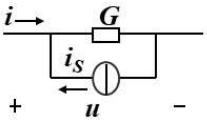
习题课

1. 求电流源发出的功率；



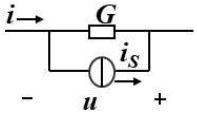
$i = Gu + i_s$

$P_{发} = -ui_s$



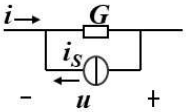
$i = Gu - i_s$

$P_{发} = ui_s$



$i = -Gu + i_s$

$P_{发} = ui_s$



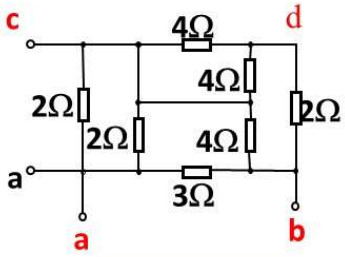
$i = -Gu - i_s$

$P_{发} = -ui_s$

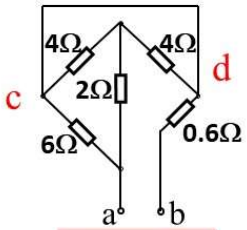
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2. 求Rab，Rac

$R_{ac}=0.833\Omega$



$R_{ab}=1.5\Omega$



$R_{ab}=3\Omega$

$R_{ab}=2.4\Omega$

